Structural analysis of the interactions network in works in a fan-fiction community

Final Dissertation

A Mobius Project

Andrea Bragante

Supervised by:

David Lainado
Cristian Consonni
Alberto Montresor

Introduction - About fanfictions and Archive of Our Own

A **fanfiction** is a fictional work written in an amateur capacity by fans

- based on an existing work of fiction.
- growing popularity with the spreading of the Web

Archive of Our Own is one of the most important fan fiction platforms.

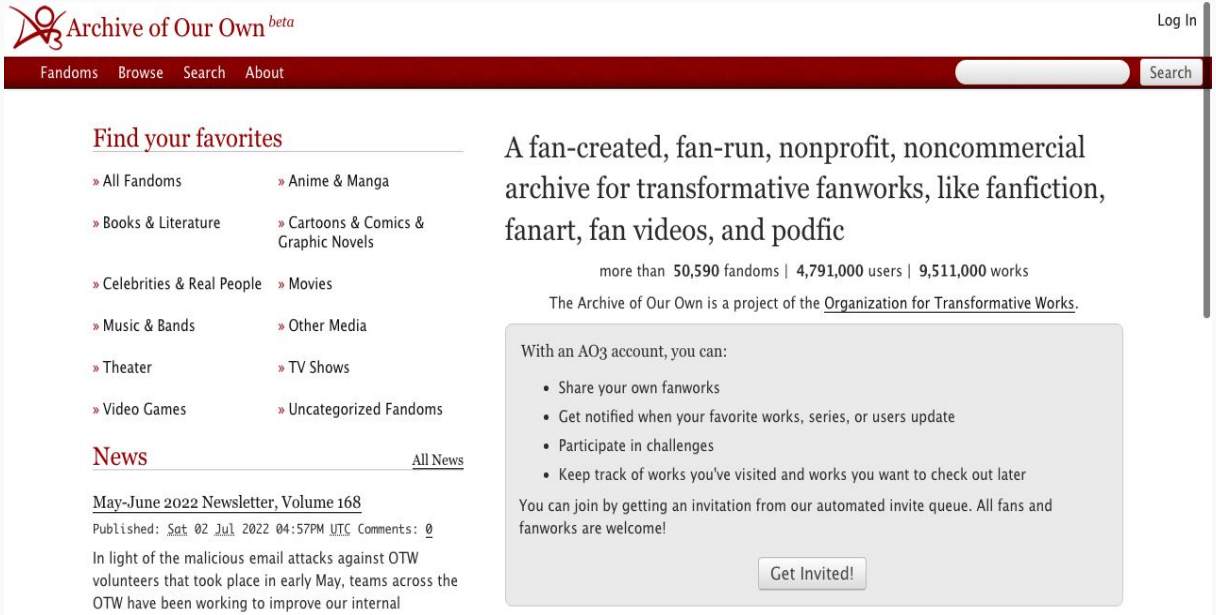
- feedback system
  - kudos
  - bookmarks
  - comments
- idea of **prosumers**: users can be at the same time *producers* and *consumers*

Community	Number of Works	Number of Comments
Marvel	444 804	10 946 620
Harry Potter	280 310	6 350 061
Sherlock Holmes	124 337	2 749 016
Lord of the Rings	24 234	289 388
Percy Jackson	19 240	311 157
Twilight	10 974	141 402
Warriors	2 763	24 472

Introduction - Research goal

The **goal** of my work was to perform an analysis on works and users of a set of fanfiction communities

- social network building
  - by users
  - by interactions (through comments)
- analyses
  - structural analysis
  - centrality analysis
  - tree analysis
  - contributions analysis



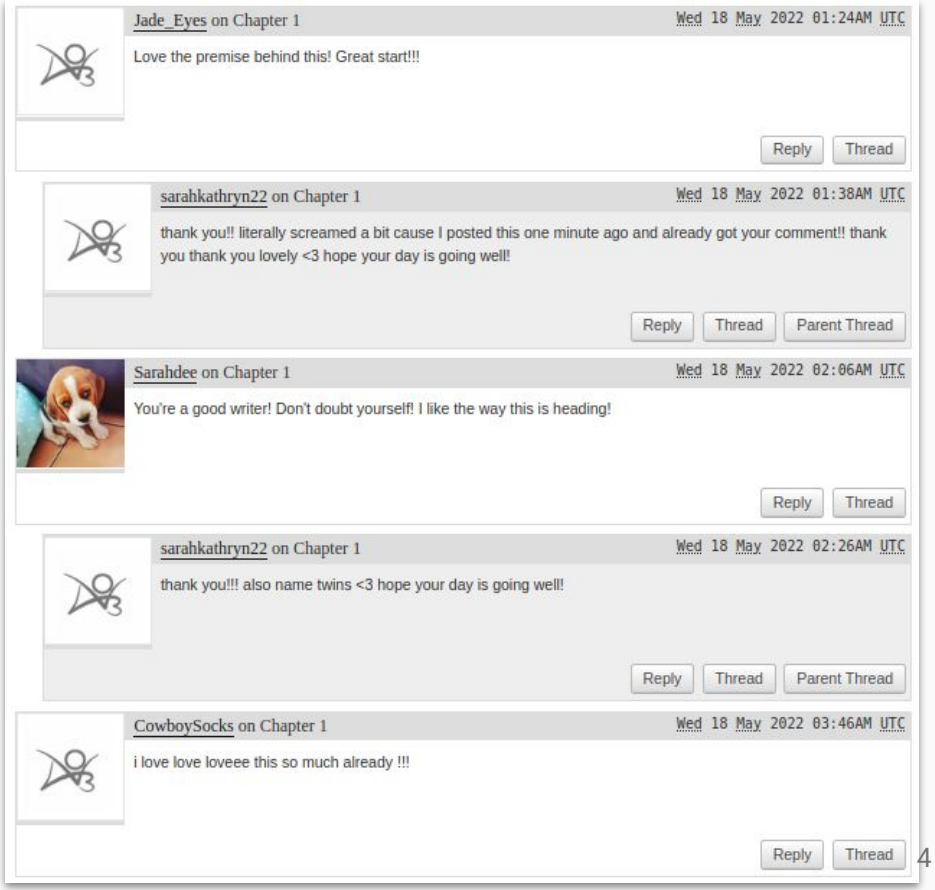
User-User interactions - Building networks from user interactions

Interaction graph

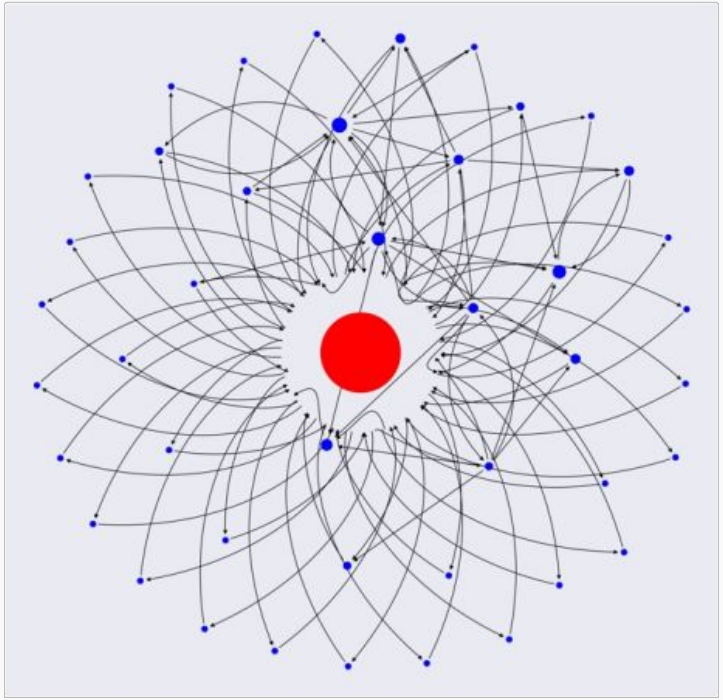
- represents the interactions between users through comments
- used to run metrics

Directed weighted graph:

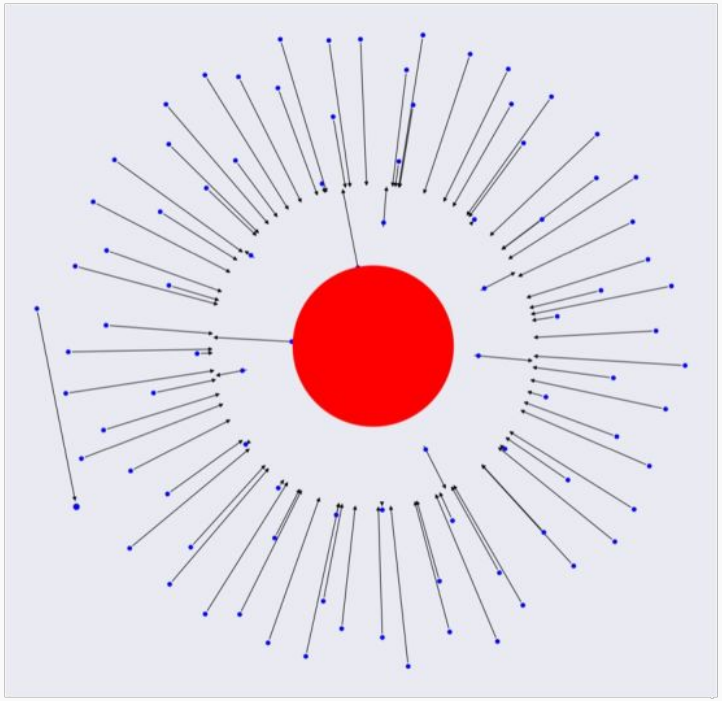
- *nodes*: users
- *edges*: interaction between two users (via comments)
- *weight*: number of replies written between the users
  - weight is not used in this analysis



User-user graphs - Work analysis



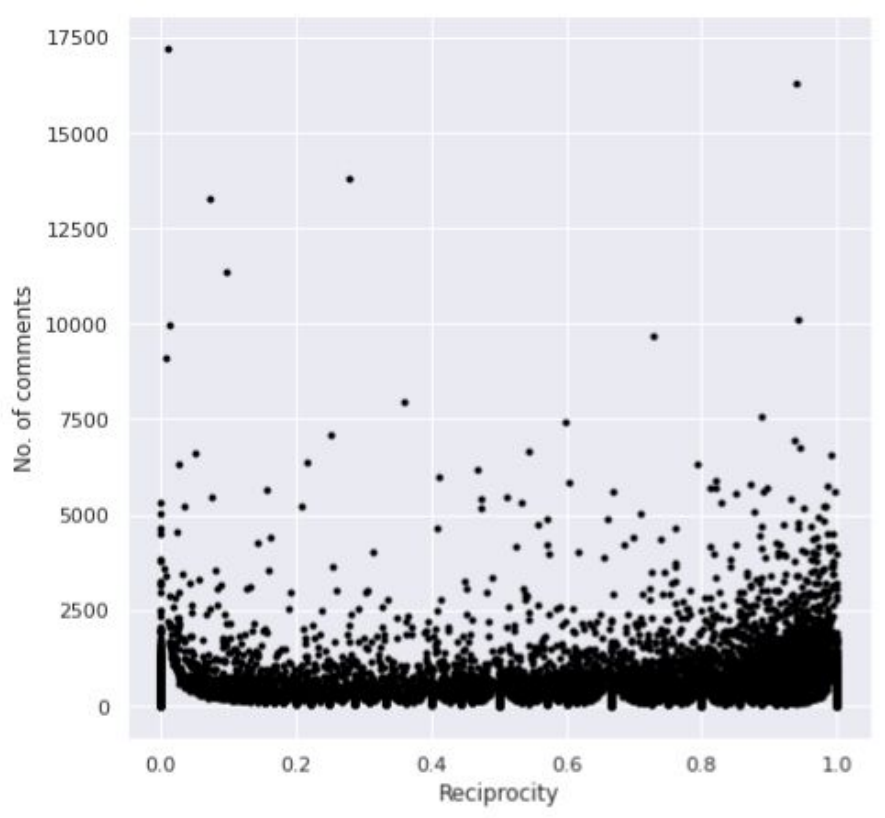
“A guy could do worse”, ***Marvel***
Users: 44, Interactions: 117
Clustering Coefficient: 0.094
Reciprocity: 0.82

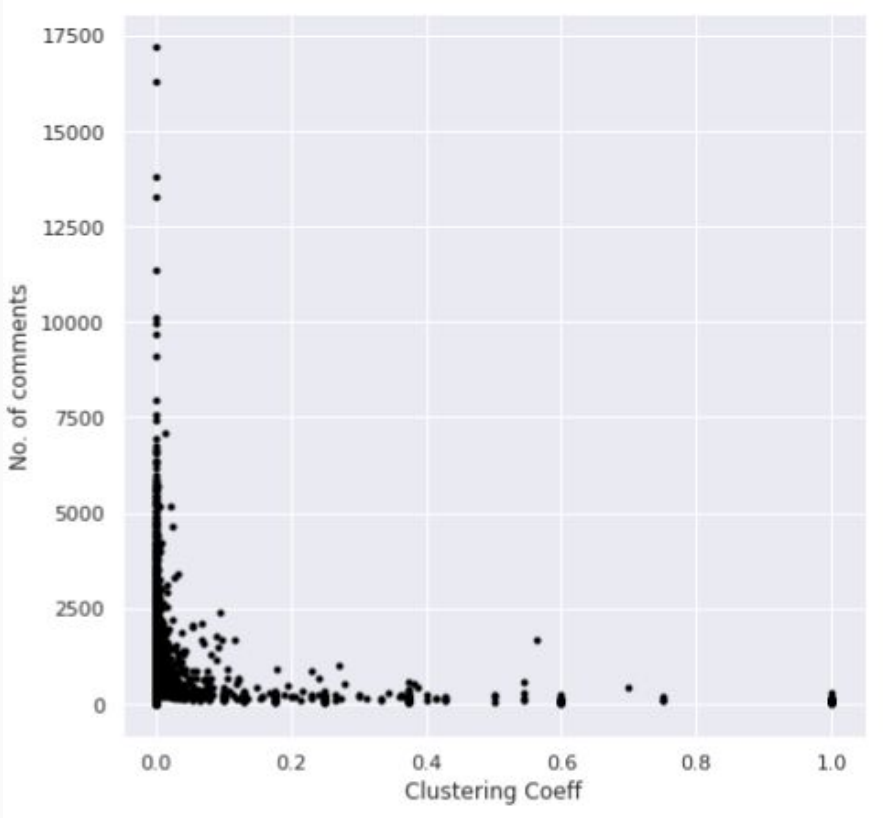


“Fairy tales and clockwork hearts”, ***Marvel***
Users: 92, Interactions: 90
Clustering Coefficient: 0
Reciprocity: 0

User-user graphs - Correlation between metrics

Correlation between **number of comments** and the structural metrics.





Comments trees - Building trees from user interactions

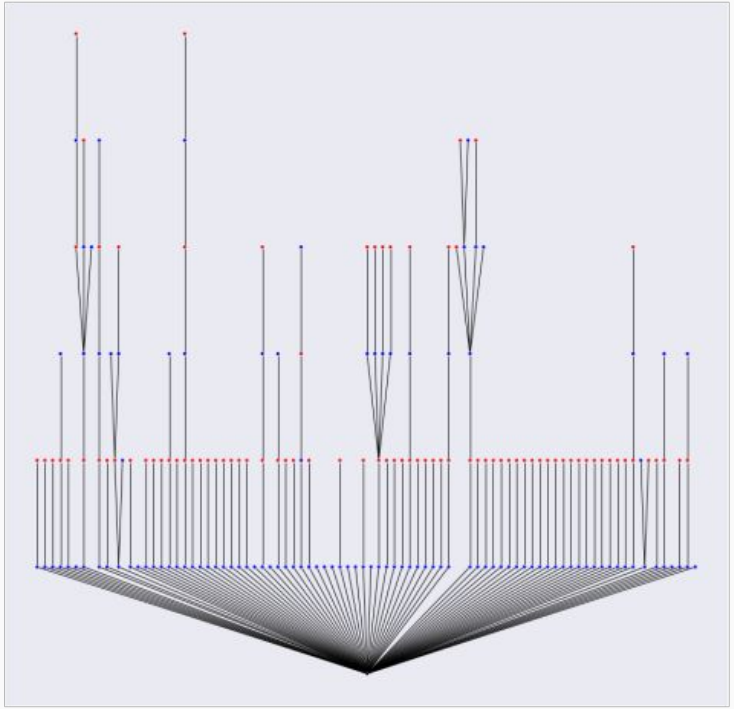
Let us consider a different kind of graph this time, a tree:

- **root** node
- each node is a **comment**
- each node *b*, child of a node *a*, is a reply comment to its parent comment *a*

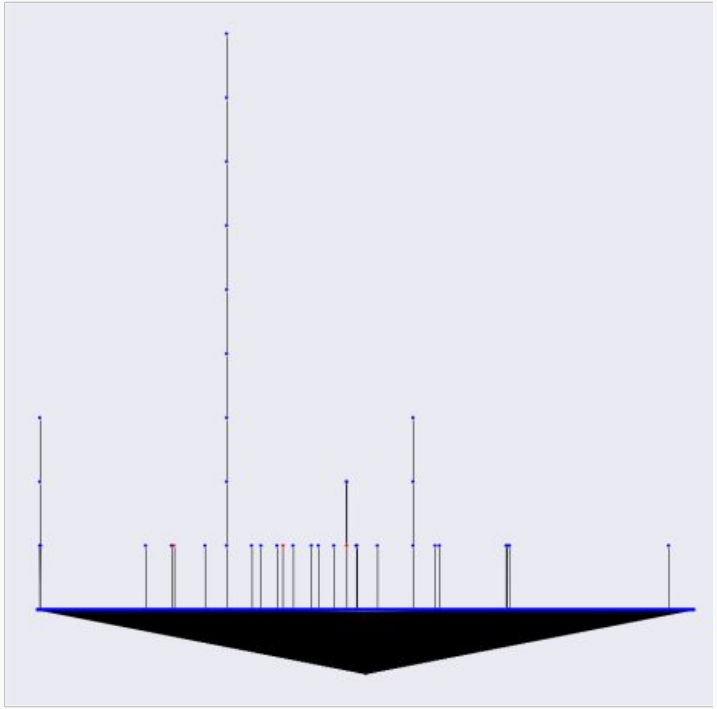
And different metrics:

- tree structure (eg. tree depth, breadth, percentage of single comments)
- author role (eg. percentage of author comments, whether the author is the most active user)
- commenters role (eg. average number of comments per user, number of distinct users)
- threads (eg. average thread length, participation in threads, percentage of threads longer than 3, 5, 10)

Comments trees - Work analysis



“Everyone in town’s mighty scared, what with all the gold and the outlaws”, ***Marvel***



“The least of all possible mistakes”, ***Sherlock Holmes***

Key points:

- first layer density
- bifurcations
- alternation of red and blue nodes

Comments trees - Aggregated work analysis

	% of **works** where the **author** is the **most active user**	Average **%** of **author comments**	Average **%** of **direct comments**	Average **thread length**	Average **%** of **threads longer than 10**
Marvel	86.1%	34.8%	25.6%	2.23	1.9%
Harry Potter	78.9%	30.8%	33.6%	2.19	1.5%
Sherlock Holmes	90.4%	37.6%	20.2%	2.27	1.8%
Lord of the rings	85.1%	36%	23%	2.37	2.9%
Percy Jackson	80.9%	31.9%	28.6%	2.36	2.3%
Twilight	65.2%	25.8%	44%	2.24	1.4%
Warriors	70.2%	31.1%	28.8%	5.27	4.4%

Users in different contexts - Building the contributions DataFrame

The questions we are trying to answer are

- how does one user's **behaviour** change when the **context** changes?
- is there such a thing as a **general attitude** when commenting works?

Let us try to analyse the **behaviour** of users via new metrics:

- comments under other people's works and own works
- average **number of comments**, **pagerank**, **in/out degree** under other people's works and own works

We choose a set of users to work with (ca. 23000)

- an author must have written **at least 5** works (marvel: 10)
    - for computational reasons

Users in different contexts - Analysing aggregated contributions DataFrame

	% of authors who wrote **no comments**	Authors **more active** under own works (**no. comments**)	Authors **more active** under own works (**pagerank**)	Authors **more active** under own works (**out-degree**)	Authors **more active** under own works (**in-degree**)
Marvel	3.1%	48.5%	54%	69.8%	72.1%
Harry Potter	9.4%	32.6%	38.7%	52.8%	55.7%
Sherlock Holmes	5.9%	38.7%	37.1%	54.6%	56.5%
Lord of the rings	12.5%	25.5%	43.3%	44.5%	45.2%
Percy Jackson	6.4%	26.9%	50.2%	51.5%	53.5%
Twilight	22.5%	9.6%	42.6%	38%	43.2%
Warriors	16.9%	18.3%	33.8%	35.2%	35.2%

Results and Conclusions

User-user graphs:

- the **author** seems to be quite often the **central node** in U-U graphs (star-topology)
- **clustering coefficient** tends to be **low** and to grow on **inverse** correlation respect to the **number of comments**

Comments trees:

- the more the **author comments**, the more the **threads grow in depth**
- there are **general trends** among all the communities

Author role in context:

- most of the authors are also active consumers (**prosumers**)
- some users **over-participate** under their own works

Thanks.